MOHAMED AMINE CHOUCHENE

Computer Science Student — Technical University of Munich

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PROFILE

Informatics student at TUM with hands-on experience in Java, C/C++, SQL, JavaScript, HTML, and CSS through academic and personal projects. Strong foundation in object-oriented programming, backend development, and database design. Looking for a Werkstudent position to apply and grow software development skills in a professional team environment.

EDUCATION

Technical University of Munich (TUM) — B.Sc. Informatics *10.2024 – Present*

Munich, Germany

- Introduction to Software Engineering: studied the full software development lifecycle — requirements analysis, conception, architecture and design, implementation, testing, analysis, deployment, and DevOps practices.
- Additional coursework: Algorithms & Data Structures, Databases, IT Security, Operating Systems, Business Analytics & Machine Learning.
- Focus on programming and software development with Java, C/C++, JavaScript, Python, and SQL.

Ben Arous High School — Baccalaureate in Computer Science *09.2019 – 07.2023*

Ben Arous, Tunisia

- Ranked 1st nationwide in the Tunisian Baccalaureate (Computer Science track); awarded a DAAD-affiliated government scholarship.
- Built foundational web development and database skills: HTML, CSS, JavaScript, PHP, SQL.

LANGUAGES

English (C1) · **German (C1)** · **Arabic (Native)** · **French (C1)**

PROJECTS

Secure Memory Unit — Low-Level Security Project *04.2025 – 07.2025*

University Team Project (team of 3)

- Developed a secure memory management system in C/C++, implementing encryption and decryption mechanisms for controlled data access.
- Applied memory management, data representation, and system-level programming concepts to address low-level security challenges.
- Delivered the project with a strong evaluation, gaining practical experience in C/C++ and collaborative software development.

Git Workflow, Memory Debugging & Containerization — Coursework Exercise *2025*

Individual Weekly Assignment

- Resolved Git branch conflicts and restructured commit history using merging, squashing, and cherry-picking to consolidate features into a shared main branch.
- Debugged memory-safety issues (buffer overflow, use-after-free) in a C application using GCC AddressSanitizer.
- Fixed a broken Dockerfile and defined a multi-service Docker Compose setup (application + PostgreSQL database) with volumes, environment variables, and networking.

REST API Development with Spring Boot — Coursework Exercise *2025*

Individual Weekly Assignment

- Implemented a full CRUD REST API (Create, Read, Update, Delete) in Spring Boot, applying a layered architecture with an MVC design pattern across client and server.
- Built asynchronous REST client requests with Spring WebFlux to communicate with a JavaFX front-end for creating, updating, deleting, and retrieving entities.
- Extended REST endpoints with query-parameter-based sorting, updating both server-side business logic and client-side requests.

SKILLS

- **Languages:** Java, C/C++, JavaScript, Python, SQL
- **Web:** HTML, CSS, PHP, JavaScript
- **Tools:** Git/GitHub, Docker, IntelliJ IDEA, VS Code, Linux
- **Concepts:** Object-Oriented Programming, Unit Testing, Relational Databases, Debugging & Version Control, REST APIs, MVC & Layered Architecture
- **Frameworks:** Spring Boot, Spring WebFlux

ACHIEVEMENTS

- DAAD Scholarship Recipient — national scholarship funded by the Tunisian government in cooperation with DAAD, awarded for outstanding performance in the Tunisian Baccalaureate (Computer Science track).